A Universal 8-Vertex Origami Torus Triangulation

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Abstract

An origami torus is a piecewise affine isometric embedding of a flat torus into $\boldsymbol{R}^3$. We prove that every flat torus is realized as an 8-vertex origami torus based on the triangulation with degree sequence $4, 6, 6, 6, 6, 6, 7, 7$. This provides an 8-vertex universal triangulation for origami tori. This is a preliminary version of the paper. Subsequent versions will have improved arguments and probably more conceptual explanations.

1 Introduction

1.1 The Main Result

A *flat torus* is a quotient of the form $T = \boldsymbol{R}^2/\Lambda$, where Λ is a lattice of translations of $\boldsymbol{R}^2$. The lattice Λ is called the *lattice* of T. A *triangulation* τ of a flat torus T is a decomposition of T into finitely many triangles, such that every two triangles are either disjoint, or share one vertex, or share one edge.

Two triangulations τ_1 and τ_2, respectively of flat tori T_1 and T_2, are *combinatorially equivalent* if there is a homeomorphism $h : T_1 \to T_2$ which affinely maps each triangle of τ_1 to a triangle of τ_2. The equivalence class of τ is called its *combinatorial type*. There are 7 different combinatorial types of 8-vertex triangulations of tori. Figure 1.1 below shows the 8-vertex triangulations $\tau(46666677)$ and $\tau(66666666)$. We are naming these triangulations by their degree sequence. In each of these cases, the degree sequence determines the combinatorial type. There are two triangulations with degree sequence 55666677 but otherwise the degree sequence determines the type of the 8-vertex triangulation.

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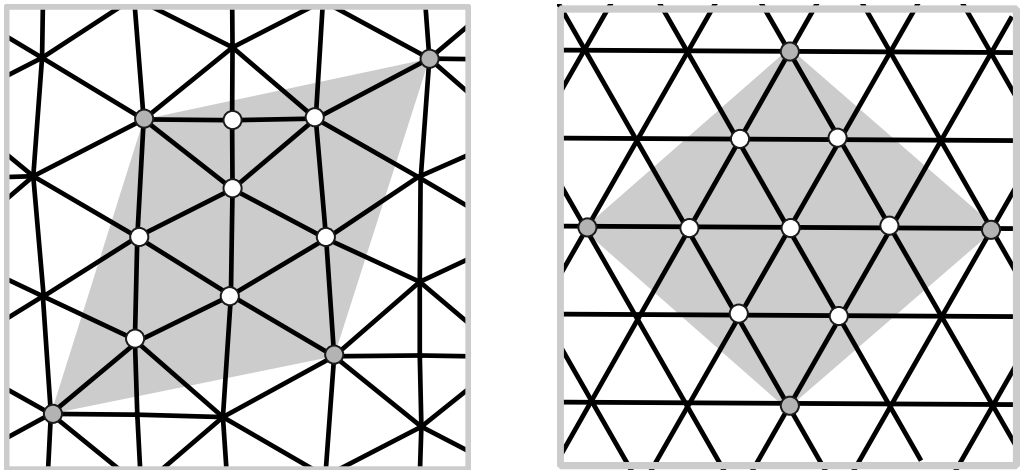


Figure 1.1: The triangulations $\tau(46666677)$ and $\tau(66666666)$.

An *origami torus* is an embedded topological torus in $\boldsymbol{R}^3$ that is made by fitting together finitely many triangles so that the cone angle around each vertex is 2π. Equivalently, an origami torus is the image of a piecewise affine isometric embedding $\phi : T \to \boldsymbol{R}^3$ of a flat torus T. This is to say that T has a triangulation τ such that the restriction of ϕ to each triangle of τ is an isometric affine map. Thus, to an origami torus, we can associate two objects:

- The combinatorial type of the triangulation τ.
- The lattice Λ.

We say that our origami torus has lattice Λ and combinatorial type τ.

We call a lattice Λ *realized* by τ if there exists an origami torus having lattice Λ and combinatorial type τ. We call a combinatorial type τ *universal* if every lattice Λ is realized by τ. Here is our main result.

Theorem 1.1 (Universality) $\tau(46666677)$ *is universal.*

The Universality Theorem says that there is a single 8-vertex combinatorial type out of which you can build an origami torus of any shape you like. One can do no better. In [**S**] it is proved that K-vertex origami tori do not exist for $K \leq 7$.

1.2 History and Context

The Universality Theorem is the culmination of a series of results about origami tori, some quite recent. In 1960, Y. Burago and V. Zalgaller [**BZ1**] give the first

construction of origami tori. In their subsequent 1995 paper [**BZ2**], they prove that one can realize every isometry class of flat torus as an origami torus. In the language above, we would say that every lattice is realized by some combinatorial type.

The works of T. Tsuboi [**T**] and (independently) P. Arnoux, S. Lelièvre, and A. Málaga [**ALM**] give an explicit construction which realizes every lattice. With minor differences, both papers realize every lattice using the origami tori known as *diplotori* [1] and first described by U. Brehm [**Br**] in 1978. See also the description of these tori in H. Segerman's book [**Se**, §6]. These constructions require infinitely many different combinatorial types of triangulation.

The 2024 paper of F. Lazarus and F. Tallerie [**LT**] gives a universal triangulation having 1217 vertices. This paper also has an excellent discussion of the various attempts made to realize all lattices as origami tori.

In [**S**], R.E.S. constructed an 8-vertex origami torus and proved that a K-vertex origami torus cannot exist for $K \leq 7$. See also [**S2**]. (There are no K-vertex triangulations with $K < 7$, so really the issue was whether a 7-vertex example could exist.) R.E.S., at the suggestion of P.G.D, called the 8-vertex origami tori in [**S**] *pup tents*, on account of their appearance. A *pup tent* is an 8-vertex origami torus T with the following features.

- The triangulation underlying T is $\tau(66666666)$.
- T has 2-fold symmetry with respect to the map $(x, y, z) \to (-x, -y, z)$.
- The origami torus has exactly 6 triangles on its convex hull boundary.

In [**DS**], R.E.S. and P.G.D. prove that almost every lattice is realized by a pup tent. More precisely, they prove that every lattice is realized provided that it is neither rectangular nor sharp rhombic. A lattice is *rectangular* if it is generated by a pair of perpendicular vectors and *sharp rhombic* if it is generated by a pair of equal-length vectors which make an angle of at most $\pi/3$ with each other. The sharp rhombic lattice generated by two equal-length vectors making an angle $\pi/3$ is especially symmetric. It is called the *hexagonal lattice*, or *hex lattice* for short. In [**DS**] the authors speculate that perhaps these kinds of lattices can be realized by 8-vertex origami tori which do not have 2-fold symmetry. The most interesting cases left open in [**DS**] are the square and hex lattices.

In the 2026 paper [**L**], F.L. proves that the square lattice is realized by $\tau(66666666)$ and the hex lattice is realized by $\tau(46666677)$. This result grew out of some experiments that F.L. and S.T. did in the summer of 2026. These authors made a poster for

[1] For the rectangular lattices one needs a slightly modified construction.

R.E.S.'s birthday conference and, in an exciting development, their software could realize some rectangular and sharp rhombic lattices.

Upon further experimentation, the folding pattern used to realize the hexagonal lattice from $\tau(46666677)$ turned out to be extremely robust. This fact led to the Universality Theorem, the result we prove in the present paper.

1.3 Proof Strategy and Philosophy

If we fix a triangulation τ, the set $O(\tau)$ of all origami tori which have τ as the underlying combinatorial type is a semi-algebraic set. This is to say that $O(\tau)$ is cut out by algebraic equalities and inequalities. The equalities govern the intrinsic flatness conditions – i.e. the 2π angle sum – around each vertex, and the inequalities govern the embedding property.

Here is how this works. The variables are

1. Two vectors X, Y generating the lattice Λ.
2. A collection of n points $p_1, ..., p_n \in \boldsymbol{R}^2$. For us, $n = 8$.
3. Points $q_1, ..., q_n \in \boldsymbol{R}^3$ realizing the vertices of a member of $O(\tau)$.

The equalities mentioned above say that each relevant distance $\|q_i - q_j\|^2$ equals an expression of the form $\|p_i' - p_j'\|^2$ where p_i' and p_j' are suitable Λ-translates of p_i and p_j, chosen so that the segment joining p_i' and p_j' is isometric to the corresponding edge of τ.

The embedding conditions can be expressed by saying that various tetrahedra formed by q_i, q_j, q_k, q_ℓ are non-degenerate and have the correct orientation. This non-degeneracy is determined by the sign of a certain determinant, a polynomial function in the coordinates involved. (Technically, one can have some degenerate tetrahedra in an embedding, but for the examples we consider, the points are all in general position.)

Ideally, we would like to get explicit coordinates for the vertices of points in $O(\tau)$. This seems quite difficult, partly because there are too many choices to make. The central idea that emerged in [**DS**] is that one should try to parametrize points in the boundary $\partial O(\tau)$. This boundary is a stratified space, and the points in it correspond to *gracefully immersed origami tori*. Such an object is a piecewise affine isometric map $\phi : T \to \boldsymbol{R}^3$ which is not an embedding but which is approximated arbitrarily closely by piecewise affine isometric embeddings.

Essentially, when we move to the boundary we allow certain faces to collapse together. Roughly speaking, the more we have collapsed the faces, the lower the

dimension of the stratum we land in. For gracefully immersed origami tori, the map ϕ is very important because it records how the collapsing is done.

The collapsing operation makes certain choices for us, and gives us a better chance of finding nice parametrizations of boundary strata. In [**DS**], the authors found a 2-dimensional stratum which was essentially a copy of a fundamental domain for the modular surface, the hyperbolic orbifold that parametrizes lattices modulo similarity. The parametrization was algebraic, and very nearly rational. That is, the variables mentioned above all had explicit algebraic expressions. Aside from one square root, the parametrization was rational. Once the authors found this parametrization, they found an algebraic path from the boundary stratum back into $O(\tau)$. We will not review this here in more detail because the strategy in the present setting is similar and easier to describe. Now we give the three step strategy for the Universality Theorem.

Step 1: In the present setting, where we work with $\tau = \tau(46666677)$, we give an explicit rational parametrization of a 5-dimensional boundary stratum. This stratum is very special: The corresponding gracefully immersed origami tori lie entirely in the plane! This is the first step of our construction. We call these origami tori *flattened origami tori*. We will give formulas and a geometric description in §2.

Step 2: We think of our origami tori as living in $\boldsymbol{R}^2 \times \{0\}$. Once we have parametrized our flattened origami tori, we will establish (to a sufficient extent) that they have a certain *lifting property*. This means that we will keep the XY-coordinates the same and add suitable Z-coordinates $(tz_0, ..., tz_7)$, depending on the example, so that the flattened origami torus becomes an embedded polyhedral torus. This embedded polyhedral torus will not quite be flat; its angle sums will equal 2π up to an error of $O(t^2)$. At the same time, the relevant determinants will increase from 0 at a rate that is linear in t. In other words, this purely vertical lifting fixes the embedding problem at a much faster rate than it messes up the flatness property. What we mean by *to a sufficient extent* is that we will not establish this lifting property for the full 5-dimensional family, but only for a 2-dimensional subset which is sufficient to realize every similarity class of lattice. We carry this out in §3.

Step 3: Our lifting path does not quite give us origami tori on the nose. To correct the lifting path, we will use a strategy similar to what is done in [**DS**]. We prove a transversality result, essentially the Implicit Function Theorem, which shows that we can recover perfect flatness by an $O(t^2)$ perturbation of the coordinates – a perturbation that is too small to mess up the embedding property. We prove this in §4.

1.4 Computer Calculations

We perform our calculations in Mathematica [**W**]. These calculations are all fairly short and easy to reproduce. One can download our code at
http://www.math.brown.edu∼res/Java/U.tar

1.5 Artificial Intelligence Use

We used Artificial Intelligence in a moderate way. The conceptual framework for the proof is ours, and evolved from the papers [**S1**], [**DS**], and [**L**]. At various stages, ChatGPT 5.6 implemented some of our search algorithms and optimized some of our code. Perhaps needless to say, the AI also helped format equations, provide references, and proofread the paper. Also, we mention that ChatGPT had an excellent idea for simplifying the angle-derivative formulas for the Jacobian calculation in §4.

1.6 Acknowledgements

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2 The Flattened Tori

2.1 The 5 Parameter Family

We are going to describe a 5-parameter family of flattened origami tori. We think of this family as a fiber bundle over a base space coordinatized by (x, y). The fiber is parametrized by variables (u, k, λ) which satisfy certain algebraic constraints.

The Lattice: The variables (x, y) correspond to a lattice

$$\Lambda = \boldsymbol{Z}(1, 0) + \boldsymbol{Z}(x, y). \tag{1}$$

That is, Λ is the $\boldsymbol{Z}$-lattice generated by $(1, 0)$ and (x, y). Here we are suppressing the dependence of Λ on x and y.

The Parameters: The base space [2] will be the closed set

$$x \in [0, 1/2], \qquad y > 0, \qquad x^2 + y^2 \geq 1. \tag{2}$$

[2]The entire base space for the family is larger, but we only care about this subset.

This set is the fundamental domain for the group we get by adjoining reflection in the Y-axis to the modular group. Up to scaling and reflection, every flat torus is parametrized by a modulus $x + iy$ in our domain. Abusing notation slightly, we call our domain the *modular fundamental domain*.

Here are the constraints on the fiber (u, k, λ).

$$u, k, \lambda/2, x + k - u \in (0, 1/2).$$

$$u < \frac{\lambda(1-\lambda)C + \lambda(x + k - 1/2)}{\lambda + 2k(1-\lambda)}, \qquad C = (1-x)(1-2k) + 2k^2. \tag{3}$$

Remarks: These constraints are hard to understand but here is some explanation. For each choice of $k, \lambda \in (0, 1/2) \times (0, 1)$ there is some interval $U_{x,k,\lambda}$, possibly empty, of choices for u. However, we will never actually need to use these constraints. Rather, we will present explicit 2-parameter subfamilies and then directly check that they have the needed properties. The complicated algebraic nature of the constraints reflects some beautiful geometry behind these families.

The Flattened Origami Torus: Here are the points of the corresponding flattened torus $T(x, y, u, k, \lambda)$. Again, we will simplify this unwieldy expression by taking explicit choices for u, k, λ as functions of x and y.

$$p_0 = (0,0,0),$$

$$p_1 = \left(u,\ \frac{u(\lambda(1-x-2k)+2k)}{\lambda y},\ 0\right),$$

$$p_2 = \left(u-\frac{1}{2},\ \frac{x(1/2-u)}{y},\ 0\right),$$

$$p_3 = \left(-k-\lambda(1-x-2k),\ \frac{\lambda(y^2-(1-x-2k)^2)-2k(1-x-2k)}{2y},\ 0\right),$$

$$p_4 = \left(x+k-\frac{1}{2},\ \frac{y^2+x(1-x-2k)}{2y},\ 0\right),$$

$$p_5 = \left(u-\frac{1}{2},\ \frac{y^2+2(1-x-2k)(u-k-x/2)}{2y},\ 0\right),$$

$$p_6 = \left(u-k,\ \frac{(1-x-2k)(u-k)}{y},\ 0\right),$$

$$p_7 = \left(u,\ \frac{u(1-x-2k)+\frac{1}{2}((1-x-2k)^2+y^2)-\frac{u-x-k+1/2}{1-\lambda}}{y},\ 0\right).$$

The Flat Torus: We will describe 16 triangles. We get a tiling of $\boldsymbol{R}^2$ by taking the orbit of these triangles under the action of the lattice Λ defined above. The quotient then gives us our 8 vertex triangulation of $\boldsymbol{R}^2/\Lambda$. The triangles are as follows:

The triangles of the fundamental domain are

$$\begin{array}{ll|ll}
012 & (w_0,w_1,w_2) & 167 & (w_1,w_6,w_7)\\
031 & (w_0,w_3,w_1) & 264 & (w_2+e,w_6,w_4+e-\tau)\\
024 & (w_0,w_2,w_4-\tau) & 257 & (w_2+e,w_5+e,w_7)\\
043 & (w_0,w_4-\tau,w_3) & 276 & (w_2+e,w_7,w_6)\\
152 & (w_1,w_5,w_2) & 347 & (w_3+\tau,w_4,w_7)\\
136 & (w_1,w_3,w_6) & 356 & (w_3,w_5+e-\tau,w_6)\\
145 & (w_1,w_4,w_5) & 375 & (w_3+\tau,w_7,w_5+e)\\
174 & (w_1,w_7,w_4) & 465 & (w_4+e-\tau,w_6,w_5+e-\tau)
\end{array} \tag{4}$$

Here $e=(1,0)$ and $\tau=(x,y)$. The points $w_0,...,w_7$ are the same as $p_0,...,p_7$,

except for w_3 and w_6:

$$w_3 = \left(k,\ -\frac{2k(1-x-2k)+\lambda((1-x-2k)^2+y^2)}{2y}\right),$$

$$w_6 = \left(u+k,\ \frac{(1-x-2k)(u-k)}{y}\right).$$

Preliminary Isometric Mapping: Let W_k denote the kth triangle above. We call the set $\{W_0, ..., W_{15}\}$ the *fundamental triangles*. Let P_k denote the corresponding triangle whose vertices are the p_i.

Lemma 2.1 *The affine map $W_k \to P_k$ is an isometry for $k = 0, ..., 15$.*

Proof: This is a direct computational proof. We carry this out in the Mathematica file **flatness_proof** ♠

Below we will establish (in special cases) that the union F of the fundamental triangles is a fundamental domain for the action of Λ. Hence F/Λ is isometric to $\boldsymbol{R}^2/\Lambda$, and the 16 fundamental triangles give a triangulation of $\boldsymbol{R}^2/\Lambda$. Call this the Fundamental Domain Lemma. Once we have the Fundamental Domain Lemma, we can promote the map in Lemma 2.1 to a piecewise affine and arc-length-preserving mapping $\phi : \boldsymbol{R}^2/\Lambda \to \boldsymbol{R}^2$. One can picture this map as a kind of piecewise isometric folding map. This promoted map ϕ is our flattened origami torus.

The constraints for our variables listed above guarantee the good triangulation result. However, we will not prove this in full generality. Rather, we will prove this statement for our four special families. Accordingly, we now define the special families and then return to Fundamental Domain Lemma.

2.2 The Special Families

First Pass: We are interested in 4 special families but first we define 2 special families and then split each family into two pieces. The first family is given by the parameter choices:

$$\text{Family 1}: \quad u = \frac{3}{128}, \quad k = \frac{11}{32}, \quad \lambda = \frac{13}{32}.$$

$$\text{Family 2}: \quad u = \frac{x}{2} - \frac{1}{2048}, \quad k = \frac{1}{16}, \quad \lambda = \frac{1}{2}.$$

These families are defined for all (x, y) in Equation 2, but may not satisfy the constraints for all choices. We do not care about this. The variable choices in both families are independent of y and the variable choices in the first family are independent of both x and y. We are interested in the first family for $x \in [0, x_0]$ and in the second family for $x \in [x_0, 1/2]$. Here

$$x_0 = f(\frac{5}{64}) = \frac{640}{4121} = .1553..., \qquad f(t) = \frac{2t}{t^2+1}.$$

With these restrictions, the constraints are satisfied. Again, however, we do not need to check this.

The value of x_0 looks like a crazy choice, but $5/64$ is a fairly simple dyadic rational and f is a natural function that arises in the rational parametrization of the unit circle. There is some amount of leeway in our choice of x_0 but the tolerance for what we need, described below, is rather small.

Second Pass: Let $\mathcal{M}$ denote the modular fundamental domain, the set of (x, y) satisfying Equation 2. Let $\mathcal{M}[a, b]$ denote the subset of $\mathcal{M}$ with $x \in [a, b]$. Now we give a parametrization of $\mathcal{M}[0, x_0]$ by two unit squares. Let

$$t_* = \frac{5}{64}t, \qquad x(t, h) = \frac{2t_*}{1+t_*^2}. \tag{5}$$

We use two parameter squares.

Square 1A: For $(t, h) \in [0, 1] \times [0, 1]$, let

$$y(t, h) = \frac{1-t_*^2}{1+t_*^2} + h. \tag{6}$$

Square 1B: For $(t, h) \in [0, 1] \times (0, 1]$, we keep $x(t, h)$ as above and we set

$$y(t, h) = \frac{1-t_*^2}{1+t_*^2} + \frac{1}{h}. \tag{7}$$

Next, we give a parametrization of a set that contains $\mathcal{M}[x_0, 1/2]$. We set

$$t_{**} = \frac{5}{64}(1-t) + \frac{18}{64}t. \tag{8}$$

Following this, we define Square 2A and Square 2B just as we defined Square 1A and Square 1B. We just replace t_* with t_{**} in the definition.

Remark: The set we parametrize contains some points (x, y) where $x > 1/2$ but such points are irrelevant for our proofs. To get an exact parametrization we would need to take $2 - \sqrt{3} = .2679...$ in place of $18/64 = .28125$.

Now we define our 4 families. We let $\Phi_{1A}(t, h)$ denote the member of family 1 corresponding to (x, y) where $x = x(t, h)$ and $y = y(t, h)$. All this is defined relative to Square 1A. We make the same definitions for Φ_{1B} and Φ_{2A} and Φ_{2B} relative to the other squares, and in the obvious way.

Pictures: Figure 2.1 shows the triangulated flat torus associated to $\Phi_{1A}(1/2, 1/2)$.

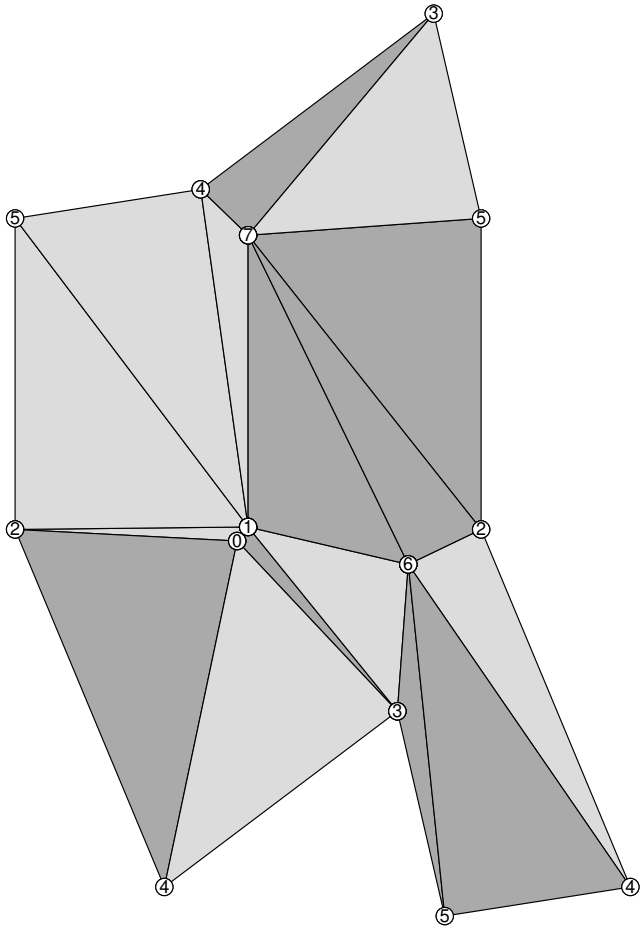


Figure 2.1: The triangulated flat torus associated $\Phi_{1A}(1/2, 1/2)$.

From the vertex labeling you can see how the action of Λ identifies the edges along the boundary to make the torus. We have shaded the triangles to indicate whether the map from these triangles to the flattened origami torus preserves or reverses orientation. Up to a global sign, the binary sequence 0110000010111101 carries all the information about the orientations.

Figure 2.2 shows the origami torus built triangle by triangle. The ordering of the faces is compatible with the ordering coming from the lift we construct in §3.

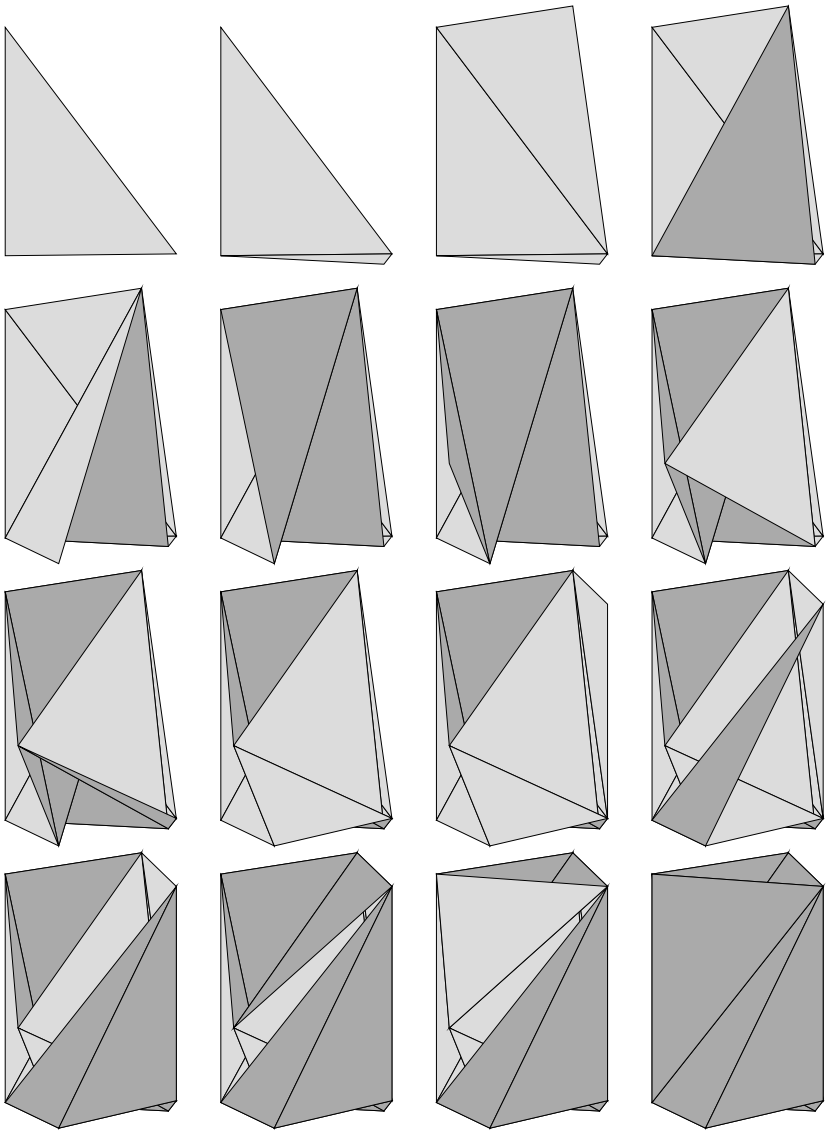


Figure 2.2: The flattened origami torus $\Phi_{1A}(1/2, 1/2)$.

The picture for $\Phi_{1B}(1/2, 1/2)$ looks similar but is much more elongated. We omit the picture, but the interested reader can draw it using our software.

Figures 2.3 and 2.4 show the same thing as Figures 2.1 and 2.2 for $\Phi_{2A}(1/2, 1/2)$.

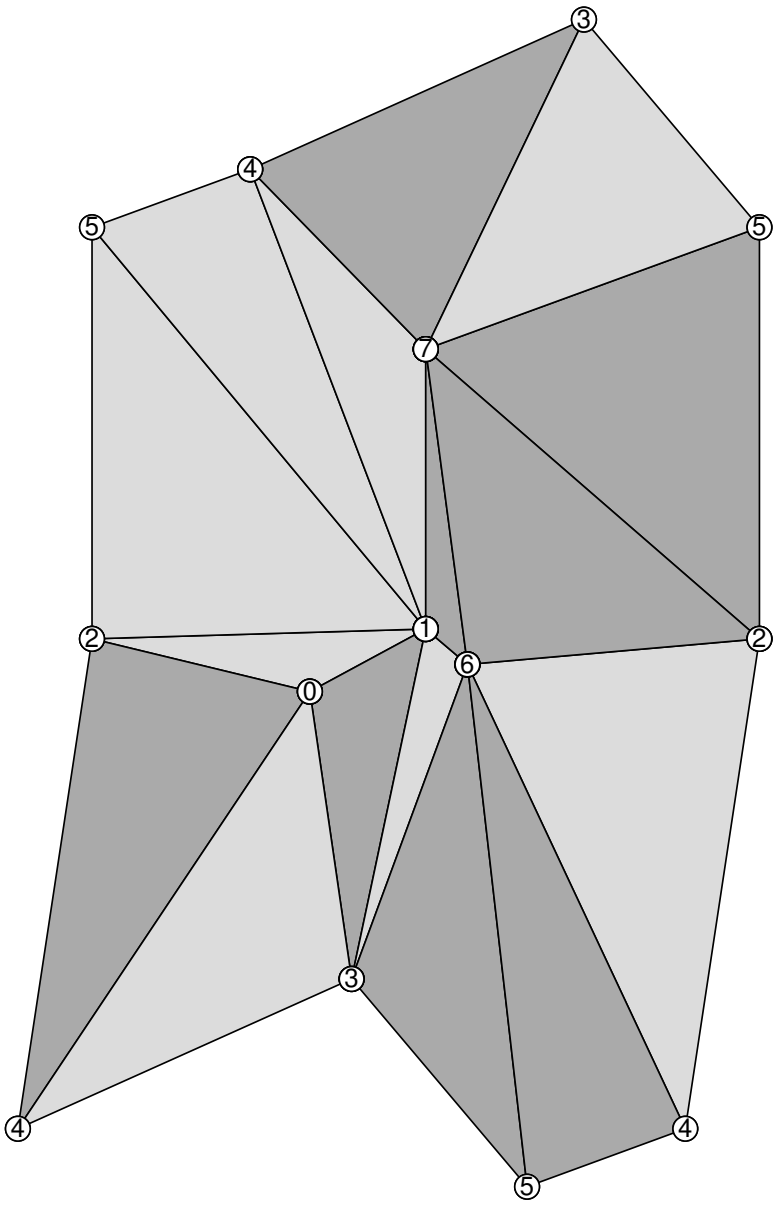


Figure 2.3 The triangulated flat torus associated to $\Phi_{2A}(1/2, 1/2)$.

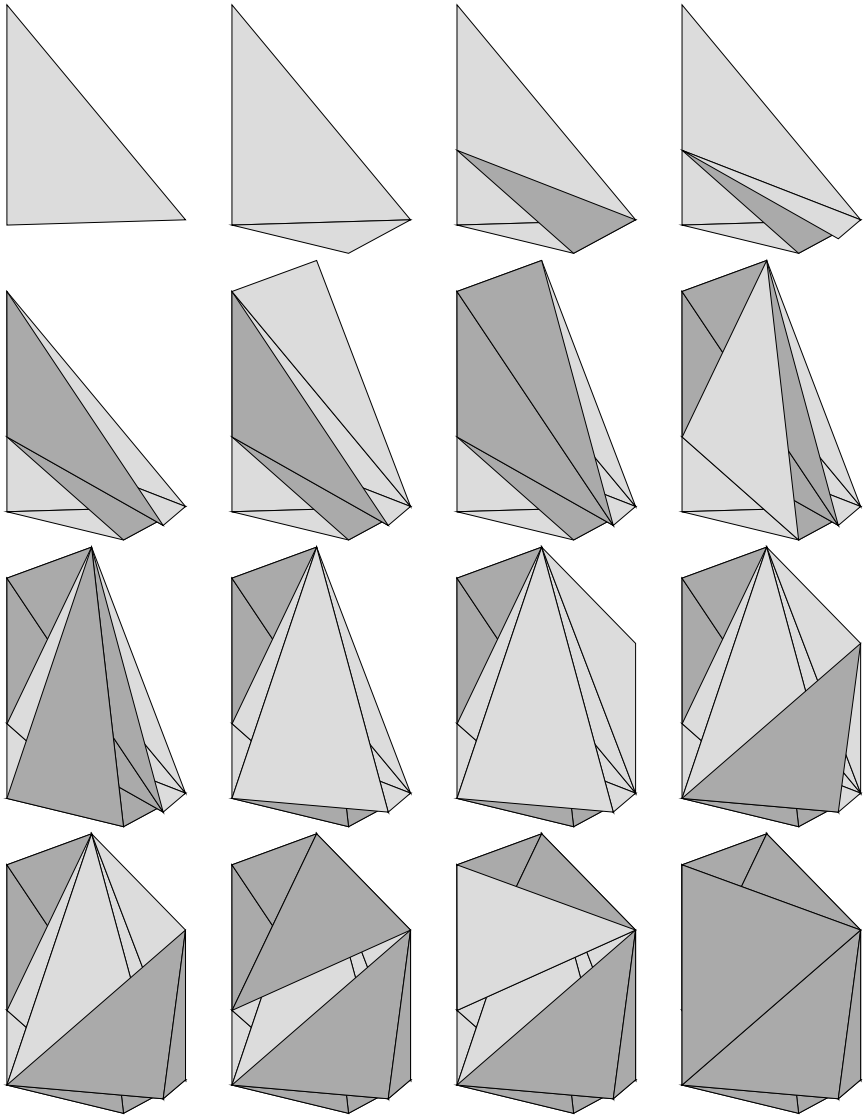


Figure 2.4: The flattened origami torus $\Phi_{2A}(1/2, 1/2)$.

Again, the picture for $\Phi_{2B}(1/2, 1/2)$ looks similar but is much more elongated.

The triangles in Figures 2.1 – 2.4 are all non-degenerate, and an inspection of the triangles in Equation 4 shows that F is a fundamental domain for Λ in these cases. As long as the triangles do not degenerate, this fact remains true for all parameters. We will reduce this non-degeneracy property to a statement that certain polynomials associated to the areas of the fundamental triangles never vanish. We first discuss a positivity criterion for polynomials on the unit square.

2.3 Positive Dominance

This material is adapted from [**S3**]. Given a polynomial $P(x) = a_0 + a_1 x + ... + a_n x^n$ define $A_k = a_0 + ... + a_k$. We say that P is *positive dominant* if all $A_k > 0$ for all k.

Lemma 2.2 *If P is positive dominant then $P > 0$ on $[0,1]$.*

Proof: If $n = 0$ or if $x = 0$ the result is obvious. Let $x \in (0,1]$ we have

$$P(x) \geq a_0 x + a_1 x + a_2 x^2 + ... + a_n x^n = x(Q(x)) > 0.$$

The final inequality comes from the fact that $Q(x)$ is again positive dominant and from induction on the degree. ♠

Now we consider a 2 variable version of this. Let $P \in \boldsymbol{R}[x_1, x_2]$ be a two-variable polynomial. We can write

$$P = \sum_I c_I X^I, \qquad X^I = \prod_{i=1}^{2} x_i^{I_i}. \tag{9}$$

Here each I has the form (I_1, I_2). Given two multi-indices I and J, we write $I \preceq J$ if $I_i \leq J_i$ for $i = 1, 2$. Define

$$P_J = \sum_{I \preceq J} c_I. \tag{10}$$

We call P *positive dominant* if $P_J > 0$ for all J.

Lemma 2.3 *If P is positive dominant, then $P > 0$ on $[0,1]^2$.*

Proof: Let

$$P = f_0 + f_1 x_2 + \cdots + f_m x_2^m,$$

where $f_0, ..., f_m$ are single-variable polynomials in x_1. Each polynomial $P_j = f_0 + \cdots + f_j$ is positive dominant. By the single-variable result, $P_j > 0$ on $[0,1]$. Now fix $x_1, x_2 \in [0,1]$. The resulting polynomial $Q(x_2)$ has successive partial sums $A_j = P_j(x_1) > 0$ for $j = 0, ..., m$. Hence $Q > 0$ on $[0,1]$ by the single-variable case. ♠

Numerator Selection: If $f = f_1/f_2$ is a finite rational function on $[0,1]^2$, written so that f_1, f_2 have no common factors, we choose f_1 so that $f_1(1/2, 1/2) > 0$. We will of course only use this selection when $f_1(1/2, 1/2) \neq 0$. If $f_1 > 0$ on $[0,1]^2$ then it follows that $f_2 \neq 0$ on $[0,1]^2$. Hence f does not vanish on $[0,1]^2$. This trick allows us to apply our positive dominance criterion to f_1 and conclude that f does not vanish.

As a variant of this phenomenon, we will consider when f_1/f_2 is finite in $D = [0,1] \times (0,1]$. Our proof that $f_1 > 0$ on D translates, in the same way as above, to the statement that f does not vanish in D.

2.4 The Fundamental Domain Lemma

Corresponding to our four families, we have 4×16 continuously varying collections of fundamental triangles. For two of these families, the triangles are parametrized by $[0,1]^2$. For the other two, they are parameterized by $[0,1] \times (0,1]$. We use h and t, as above, for the parameters.

Lemma 2.4 *At every parameter, every relevant fundamental triangle has positive area.*

Proof: Each triangle T has vertices A, B, C which are rational functions of the variables t, h. We compute the rational function

$$\Delta_T = \det(A - C, B - C),$$

and then select the numerator as in the previous section. In the case of the Families 1B and 2B we divide by h if the numerator has a factor of h in it. This happens once for each family.

Having made this small modification for two of 64 polynomials we get, we then check that all 64 polynomials are positive dominant. We use the Mathematica file **triang_proof** to do the calculation. This shows that all the areas are positive. ♠

Lemma 2.5 (Fundamental Domain) *For each parameter of each family the union F of fundamental triangles is a fundamental domain for the action of Λ, and the quotient F/Λ is isometric to $\boldsymbol{R}^2/\Lambda$.*

Proof: Our proof uses the familiar property that in a connected topological space X, a nonempty subset $S \subset X$ equals X provided that S is open and closed. In all cases X is the unit square, either with or without its bottom boundary.

Consider Family 1A. Let $S \subset [0,1]^2$ denote the set of parameters for which the Fundamental Domain Lemma is true. Given the combinatorial action of Λ on the fundamental triangles, S is an open set. Lemma 2.4 implies that S is closed because the fundamental triangles cannot degenerate at any point of $[0,1]^2$. Looking at the pattern of triangles in Equation 4 and comparing it with Figure 2.1, we see that S contains the parameter $(1/2, 1/2)$. This is a crude check which just involves the rough positions and orientations of the triangles. Hence, for Family 1A, we have $S = [0,1]^2$.

Consider Family 1B. Again, the set S is open and closed. The Family 1A tori on the top edge of $[0,1]^2$ coincide as a set with the Family 1B tori on the top edge of $[0,1] \times (0,1]$. Hence, in this case S is nonempty as well. Hence $S = [0,1] \times (0,1]$.

The analysis for Families 2A and 2B works the same way.

We mention finally that the non-degeneracy of the triangles is sufficient because their orientations are constant throughout each family. Thus, no overlaps can develop as we move around in the parameter space. ♠

Again, the Fundamental Domain Lemma lets us promote our map above to a flattened origami torus $\phi : \boldsymbol{R}^2/\Lambda \to \boldsymbol{R}^2$.

3 The Lifting Property

3.1 The Main Result

We think of $\boldsymbol{R}^2$ as the subset $\boldsymbol{R}^2 \times \{0\} \subset \boldsymbol{R}^3$. Let $\pi : \boldsymbol{R}^3 \to \boldsymbol{R}^2$ be the projection map. Let $\phi : T \to \boldsymbol{R}^2$ be a flattened origami torus. We say that a *lift* of ϕ is a piecewise affine embedding $\widehat{\phi} : T \to \boldsymbol{R}^3$ such that $\pi \circ \widehat{\phi} = \phi$. The data for a lift simply amounts to choosing a list $(z_0, ..., z_7)$ of third coordinates for the 8 vertices of $\phi(T)$. In this chapter we prove the following result.

Theorem 3.1 *Every torus in each of the four special families has a lift.*

Before we prove this result, let's explore its consequences. If $(z_0, ..., z_7)$ is a lift of ϕ then so is $(tz_0, ..., tz_7)$ for any $t > 0$. Thus, a single lift gives us a path of lifts, which we denote by $\widehat{\phi}_t$. Thanks to the Pythagorean Theorem, the edges of the triangulation of $\widehat{\phi}_t(T)$ are changing by $O(t^2)$. At the same time, the robustness of the embedding is changing linearly. This will put us in a good position to apply a transversality result to finish the proof of the Universality Theorem.

Our lifts are explicit. There is one choice for families 1A and 1B and then there is another choice for families 2A and 2B. What we do is make a choice for Family 1 and Family 2, the two larger families we defined in our First Pass in §2.2. Then we get the lifts for Families 1A, 1B, 2A, 2B by composing with our rational parametrizations of strips of the modular domain $\mathcal{M}$.

Here is the lift for Family 1.

$$0 \quad 0 \quad -2 \quad 3 \quad 0 \quad -2 \quad 2 \quad 1024 \tag{11}$$

This lift is constant, and it transfers over to Families 1A and 1B in the most straightforward way.

Here is the lift for Family 2.

$$0 \quad 0 \quad -4 \quad -2 \quad 2048 \quad -4 \quad 1816x - 229 \quad 16384 \tag{12}$$

One of the coordinates here depends on x, and when we transfer this over to Families 2A and 2B we use $x = 2t_{**}/(t_{**}^2 + 1)$, where t_{**} is as in Equation 8.

Remark: These lifts are extremely delicate, and slight changes to the parameters cause failures of the embedding property. For instance, if we change 229 to either 228 or 230 we destroy the embedding property for some examples.

3.2 The Embedding Test

Let T be one of our triangulated flat tori. Let $\mathcal{T}$ be the triangulation. Let $f : T \to \boldsymbol{R}^3$ be a piecewise affine map. We assume that f is injective on each triangle separately. Here we describe how to test that f is an embedding. The property we need is as follows: for every pair τ_1, τ_2 of triangles of $\mathcal{T}$,

$$f(\tau_1) \cap f(\tau_2) = f(\tau_1 \cap \tau_2). \tag{13}$$

We *can* test for this property but this test is somewhat redundant. We prefer to have a more streamlined test.

Lemma 3.2 *Suppose that $f(e) \cap f(\tau) = \emptyset$ provided that e is an edge of $\mathcal{T}$ and τ is a triangle of $\mathcal{T}$ and $e \cap \tau = \emptyset$. Then f is an embedding.*

Proof: We will assume that this property is true and prove that f is an embedding. Suppose that f is not an embedding. Then Equation 13 fails for some pair τ_1, τ_2 of triangles.

If τ_1 and τ_2 share an edge e it means that f folds these triangles on top of each other, at least in a neighborhood of e. After relabeling we can say that the image of at least one edge $e_1 \neq e$ of τ_1 intersects $f(\tau_2)$ in a nontrivial segment. But then let τ_1' be the triangle adjacent to τ_1 across e_1. Equation 13 also fails for (τ_1', τ_2) and these triangles do not have an edge in common. (For this claim, we use the fact that all vertex degrees are at least 4.) So, we need only worry about pairs of triangles which do not share a common edge.

If $\tau_1 \cap \tau_2 = \emptyset$ and $f(\tau_1)$ intersects $f(\tau_2)$ then some edge of one of these triangles intersects the other triangle. So, we only have to worry about Equation 13 when τ_1 and τ_2 intersect in one vertex.

Suppose then that Equation 13 fails for a pair of triangles which intersect in one vertex v. We apply an invertible affine map so that $f(v)$ is the origin and $f(\tau_1)$ is the triangle with vertices $(0,0,0)$ and $(1,0,0)$ and $(0,1,0)$.

Let e_1 and e_2 respectively denote the edges of τ_1 and τ_2 opposite the vertex v. Note that $f(e_2)$ must intersect the positive quadrant in $\boldsymbol{R}^2 \times \{0\}$. If $f(e_2)$ intersects a point of $f(\tau_1)$ we have a contradiction. But this means that $f(e_2)$ intersects the positive quadrant at a point separated from $(0,0,0)$ by $f(e_1)$. But then $f(e_1)$ intersects $f(\tau_2)$ and we have a contradiction. ♠

Now we turn to the question of how we tell when a line segment $(A_0, A_1) \subset \boldsymbol{R}^3$ is disjoint from a triangle $(A_2, A_3, A_4) \subset \boldsymbol{R}^3$. Here we are naming these objects by their vertices. If we delete A_k from the list we get a (possibly degenerate) tetrahedron T_k, and we let V_k denote the volume of T_k. To be precise, we put the points of T_k in increasing order and then take the signed volume. The sign counts the orientation.

Lemma 3.3 (01) *and* (234) *are disjoint provided at least one of the following products is positive:* V_0V_1 *or* V_2V_3 *or* V_3V_4.

Proof: To reason about these quantities we let $[abcd]$ denote the signed volume of the tetrahedron with vertices listed in the order a, b, c, d. For instance $V_0 = [1234]$.

If $[0234]$ and $[1234]$ have the same sign then (01) lies strictly on one side of the plane containing (234). This gives disjointness.

If (01) intersects (234) then the three volumes (0123) and (0134) and (0142) either have the same sign or at least one vanishes. Hence if (0123) and (0142) have opposite signs we have disjointness. Since $(0142) = -(0124)$ this is the same criterion as $V_3V_4 > 0$. The same argument applies to V_2V_3. ♠

It turns out that there are 138 pairs (e, τ) where e is an edge of $\mathcal{T}$ and τ is a triangle of T and $e \cap \tau = \emptyset$. We call these *relevant pairs*. We say that a *winning index* for a relevant pair is a choice $i \in \{0,1,2\}$ such that the ith inequality in Lemma 3.3 is true. We say that a *winning sequence* is a sequence of 138 winning indices, one per relevant pair. Combining the results from the previous section, we see that the lifted map $\widehat{\phi}$ is an embedding provided that $\widehat{\phi}(T)$ has a winning sequence.

3.3 Nonvanishing of Volumes

Here is our main computational result.

Lemma 3.4 *The following is true:*

- *All Family 1A polynomials are positive on* $[0,1]^2$.
- *All Family 2A polynomials are positive on* $[0,1]^2$.
- *All Family 1B polynomials except* F_{0167} *are positive on* $[0,1] \times (0,1]$.
- *All Family 2B polynomials except* F_{0267} *are positive on* $[0,1] \times (0,1]$.

Proof: For each quadruple $q = a,b,c,d$ and each special family we associate a polynomial $F_q(t,h)$ by computing

$$F_q = \det(p_a - p_d, p_b - p_d, p_c - p_d) \tag{14}$$

and then selecting the numerator as in §2.3. Recall here that we take the numerators to be positive at the point $(1/2,1/2)$. In this formula we have suppressed the dependence on the family and on t,h. There are 70 distinct quadruples (listed in increasing order). Thus we have 70 polynomials associated to each family.

For polynomials associated to Families 1B and 2B we factor out h whenever possible. This only happens several times in each family. (Again, this does not affect the conclusion of the argument because $h > 0$ for these families.) In the file **lift_proof** we prove the following result.

- All Family 1A polynomials except F_{0245} are EPD.
- All Family 2A polynomials are EPD.
- All Family 1B polynomials except F_{0245} and F_{0167} are EPD.
- All Family 2B polynomials except F_{0267} are EPD.

In Familiess 1A and 1B, the polynomial F_{0245} factors as $g(t)\Gamma(t,h)$ where

$$g(t) = 4096 - 4096t + 25t^2$$

and $\Gamma(t,h)$, a polynomial that depends on the family, is positive dominant. Since $g(t) > 0$ for $t \in [0,1]$ we see that both versions of F_{0245} are positive on their domains. ♠

Remark: In Families 1B and 2B, the excepted polynomials really do vanish in their domains. This is not just an artifact of our criterion.

3.4 Winning Sequences

We keep the notation from above. Suppose that the lift $\widehat{\phi}(T)$ has a winning sequence. We say that an increasing quadruple $a < b < c < d$ is *used* (relative to this winning sequence) if the associated volume participates in any of the inequalities associated to the various winning indices. Otherwise we call the quadruple *unused*.

Lemma 3.5 *In each of the four special families, the lifted torus associated to the parameter* $(1/2, 1/2)$ *has a winning sequence. In Family 1B the winning sequence* $0, 1, 6, 7$ *as an unused quadruple. In the 2B family the winning sequence has* $0, 2, 6, 7$ *as an unused quadruple.*

Proof: We check this by direct calculation. Since we have already determined the non-vanishing of the relevant volumes, our calculation just amounts to checking the signs. We do this in floating point arithmetic because we only need 2 digits of accuracy to make the check. The minimum absolute value of the determinant, taken over all used determinants in the four examples, is 0.023... (If we compute for all 70 determinants the minimum is about .0096.) ♠

Proof of Theorem 3.1: For each of the families, the same winning sequence works for all parameters because none of the functions associated to the used quadruples changes sign. So, the same logical sentence works at each parameter value. ♠

4 The Mapping Result

4.1 Lengthening a Segment

In this chapter we finish the proof the Universality Theorem. We begin by discussing a Euclidean geometry phenomenon which plays an important role in the proof.

Let $S \subset \boldsymbol{R}^2 \times \{0\}$ be a line segment of unit length. Suppose that the endpoints of S are $(0, 0, 0)$ and $(1, 0, 0)$. How long is the segment with endpoints $(0, 0, 0)$ and $(1, 0, t)$? By the Pythagorean Theorem, the length is $\sqrt{1 + t^2} \leq 1 + t^2$. This is not the best estimate, of course, but it suffices for our purposes.

The estimate (and scaling) implies that if we consider a path of lifts of a line segment into $\boldsymbol{R}^3$ by a change of height that is linear in t, then the length grows on the order of $O(t^2)$. Let us apply this principle to the lifts of our flattened origami

tori. Given a flattened origami torus $\phi : T \to \boldsymbol{R}^2 \times \{0\}$ we constructed an embedded lift

$$\widehat{\phi}_t : T \to \boldsymbol{R}^3$$

in which the heights change linearly with t. The edge lengths of the image

$$\widehat{\Psi}_t = \widehat{\phi}_t(T)$$

change by $O(t^2)$ along this path.

The image $\widehat{\Psi}_t$ is an embedded polyhedral torus. It is not quite an origami torus but it is very close. Let θ_t denote the 8-tuple of angles of $\widehat{\Psi}_t$. By elementary Euclidean geometry – e.g. the Law of Cosines the components of θ_t vary by $O(t^2)$. As we mentioned in the previous chapter, $\widehat{\Psi}_t$ becomes embedded at a linear rate whereas it loses the flatness property at the much slower rate of $O(t^2)$.

4.2 The Pseudo Modulus Function

The polyhedral torus $\widehat{\Psi}_t$ is not in general an origami torus. However, we would still like a way to assign something like a modulus to it in a meaningful way. What we want is that the modulus should vary smoothly with the vertices and agree with the modulus of the flat structure in case we have an origami torus. We construct such a function in this section. The construction depends on some choices, but these choices do not matter for the proof. Any other set of choices would work equally well.

We have been taking our moduli of flat tori to lie in the modular domain $\mathcal{M}$ defined in Equation 2. However (working with complex numbers for convenience) any complex number $\tau \in \boldsymbol{H}^2$ that is $PSL_2(\boldsymbol{Z})$ equivalent to the one in $\mathcal{M}$ would equally well serve as a modulus. Allowing τ to vary in all of $\boldsymbol{H}^2$ is the usual set up we have when we work with Teichmuller space rather than with moduli space. This is the perspective we take here.

A *marked flat torus* is a flat torus together with a distinguished basepoint and a distinguished pair of loops generating the fundamental group. Since our flat tori come with triangulations which vary continuously within a family, we can choose our markings once and for all so that the distinguished vertex is (say) the point w_0 and the two generating loops are edge paths in the triangulation. The nature of these edge paths is not important to us. The only thing that is important is that they vary continuously within a family.

In case we have a piecewise affine map $\phi : T \to \Psi \subset \boldsymbol{R}^3$ we can extract a planar parallelogram, as follows. We develop Ψ into the plane by unrolling it along the two loops. Each loop develops into a polygonal path and defines the vector that points from the initial point to the final point. Figure 4.1 shows what we mean.

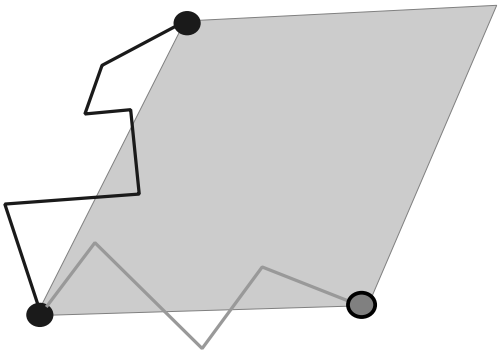


Figure 4.1: The developed parallelogram

Thus, we can assign a "modulus" τ to the triple (ϕ, T, Ψ) even when ϕ is not an isometric embedding, or even an embedding. In case ϕ is an isometric embedding, a.k.a. an origami torus, τ equals the modulus of the flat torus up to $PSL_2(\boldsymbol{Z})$ equivalence.

We also have an intermediate situation, namely when ϕ is a piecewise affine map but $\Psi = \phi(T)$ is an origami torus. In this situation, there is some other flat torus T_* with respect to which Ψ is the image of a piecewise affine isometry. We will not need to construct T_* explicitly. In the case τ measures the flat modulus of Ψ and T_*, not the modulus of T.

We call τ the *pseudo modulus function*. The key point is that τ is defined for general piecewise affine maps and it measures the modulus up to $PSL_2(\boldsymbol{Z})$ equivalence for origami tori.

4.3 The Main Calculation

Let $\phi : T \to \Psi \subset \boldsymbol{R}^2 \times \{0\}$ be one of our flattened origami tori. We set $\Psi = \phi(T)$. Given a 7-vector $\epsilon = (\epsilon_0, ..., \epsilon_6) \in \boldsymbol{R}^7$ let $\Psi(\epsilon)$ be the result of replacing the vertex (x_i, y_i) of Ψ by $(x_i + \epsilon_i, y_i)$. We do not touch vertex 7, the last vertex. We let $\theta = (\theta_0, ..., \theta_6)$ denote the 7-vector of cone angles of the perturbed triangulated torus $\Psi(\epsilon)$. We do not consider the last cone angle θ_7.

For each of the 4 families we have the Jacobian

$$J = \det \left. \frac{\partial \theta}{\partial \epsilon} \right|_{\epsilon=0}. \tag{15}$$

In other words, we take the 7×7 matrix of partial derivatives of the function $\epsilon \to \theta$ and then we take the determinant. The domain of J is $[0,1]^2$ for Families 1A and 2A. The domain of J is $[0,1] \times (0,1]$ for Families 1B and 2B.

Lemma 4.1 *For each of the four families, J does not vanish on its domain.*

Proof: We do this using the file **trans_proof**. We compute J in each case, factor the numerator, then make each factor positive in the sense of §2.3. (The denominators never vanish because the triangles are all nondegenerate.) With one exception, we check that every factor in sight is positive dominant. The one exception is just the polynomial $f(t,h) = h$, which appears in some factorizations of the versions of J associated to the B families. This polynomial is obviously positive on $[0,1] \times (0,1]$. ♠

Our calculation is straightforward except for one fine point which we discuss. The formula for the angle α between two vectors $V = (v_1, v_2)$ and $W = (w_1, w_2)$ is

$$\alpha = \arccos\left(\frac{V.W}{\|V\|\|W\|}\right).$$

This formula is far from rational, but for the derivatives we have e.g.

$$\frac{\partial \alpha}{\partial v_1} = \sigma \frac{v_2}{v_1^2 + v_2^2}, \qquad \sigma = \text{sign}(\det(V,W)).$$

Here is the crucial observation. Since the triangles in our families are non-degenerate everywhere, we can determine all the σ values once and for all by looking at a single example. (All four families have the same sequence of signs, so we only need to make one check.) This is what we do. This gives us fairly simple rational expressions for J in all four cases.

4.4 The Local Diffeomorphism

Now we make a crucial observation. For any point ζ in any of our four domains, the corresponding special family is defined in an open neighborhood of ζ. All the formulas are well-defined and hold, as long as we keep inside a sufficiently small neighborhood of ζ. If ζ is in the boundary of our domain then the modulus $\tau(t,h)$ corresponding to the lattice may lie outside the modular domain $\mathcal{M}$. However, we simply interpret τ as a point in the upper half plane $\boldsymbol{H}^2$, the Teichmuller space of marked flat tori.

All of our positivity results hold by continuity in a sufficiently small open neighborhood around ζ. Thus we have flattened origami tori for all such points and also embedded lifts and also transversality. In short, all our results hold in the neighborhood of a sufficiently small closed disk Δ centered at ζ.

We define a map

$$F : \boldsymbol{R}^7 \times \boldsymbol{R} \times \Delta \to \boldsymbol{R}^7 \times \boldsymbol{R} \times \boldsymbol{C}. \tag{16}$$

The variables for the domain are the perturbation vector ϵ discussed above, a real parameter t, and a point $p \in \Delta$.

Now we construct the map F. We let $\widehat{\Psi}_{t,p} = \widehat{\phi}_t(T_p)$ be our lifted polyhedral torus, where we lift by the vector $(tz_0, ..., tz_7)$ of heights. Finally, we perturb the X-coordinates to get $\widehat{\Psi}_{t,p}(\epsilon)$. This is our X-perturbed lifted polyhedral torus. We define

$$F(\epsilon, t, p) = (\theta, t, \tau) \tag{17}$$

Here τ is the pseudo-modulus of $\widehat{\Psi}_{t,p}(\epsilon)$. Everything in sight varies smoothly.

Lemma 4.2 *F is a local diffeomorphism at each point $(\overrightarrow{0}, 0, p)$*

Proof: By the Inverse Function Theorem, it suffices to show that the differential dF is nonsingular at this point. In our calculation we treat the real and imaginary part of the pseudo modulus τ as two real variables, τ_1 and τ_2. Thus we have a 10×10 matrix. To fix conventions, the columns are what we get by differentiating with respect to a fixed variable. The rows are what we get by differentiating a fixed variable with respect to all the others.

We now analyze the structure of dF and prove that it has the form

$$dF = \begin{pmatrix} A & 0 & 0 \\ 0 & 1 & 0 \\ B & 0 & C \end{pmatrix} \tag{18}$$

Here, the matrix A is the upper 7×7 minor of dF. This is the invertible matrix $\partial\theta/\partial\epsilon$ which we considered in our main calculation, The matrix C is an invertible 2×2 matrix. So, to finish the proof, we just have to establish Equation 18 and the invertibility of C.

We have already discussed the block A in Equation 18. Here is the rest of the analysis.

- We have $\partial\theta/\partial t = 0$ because these angles are changing by $O(t^2)$. At the same time we have $\partial t/\partial v = 0$ for any variable $v \neq t$. Thus, the upper 8×8 minor of dF has block diagonal form with a $1 = \partial t/\partial t$ on the diagonal.

- More strongly, the 8th row, representing $\partial t/\partial v$, has all zeros except for the single 1 we have already mentioned.

- The 8th column has just a single 1. The reason is that $\partial\theta/\partial t = 0$ by construction and $\partial\tau/\partial t = 0$ because all the edge lengths of our triangulation change by $O(t^2)$ and hence so does τ.
- We don't care about the bottom left 7×2 matrix B. This represents $\partial\tau/\partial\epsilon$.
- We have $\partial\theta/\partial p = 0$ because we are evaluating this entirely on flattened origami tori. This makes the upper right 2×7 matrix the 0-matrix.
- The matrix $C = \partial\tau/\partial p$. In computing these partial derivatives we are working inside the space of flattened origami tori. Recall that p is the parameter of the domain for our special families and $\tau = \tau(p)$ is $PSL_2(\boldsymbol{Z})$ equivalent to the image of the (expanded) domain into $\boldsymbol{H}^2$ given by our rational parametrization of the relevant subset of $\boldsymbol{H}^2$. Thus the map $p \to \tau(p)$ is a diffeomorphism.

This completes the derivation of Equation 18. ♠

4.5 Putting it Together

Let

$$\vartheta = (2\pi, ..., 2\pi). \tag{19}$$

This vector has length 7. Let D be an extremely tiny disk centered at the point $\tau(0, 0, \zeta)$, the point which represents the modulus of the flattened origami torus corresponding to our point ζ. Consider the set

$$\Delta_t = F^{-1}(\vartheta \times \{t\} \times D). \tag{20}$$

Here we mean to use the local inverse of F guaranteed by the Inverse Function Theorem, not the full pre-image.

Each point of Δ_t has cone angle 2π and thus is flat. (The fact that $\theta_7 = 2\pi$ follows from the Gauss-Bonnet Theorem.)

Lemma 4.3 *For $t > 0$ sufficiently small, each point of Δ_t is also embedded.*

Proof: Our original lift (which keeps $\epsilon = 0$) changes the cone angles by $O(t^2)$, Thus we only need to add a correction of $O(t^2)$ to restore the flatness. Since the various volumes which determine the embedding grow linearly with t, they remain nonzero for all points of Δ_t once t is sufficiently small. We can make a uniform argument like

this for all points of Δ_t by smoothness and compactness. ♠

In short, Δ_t consists entirely of origami tori for $t > 0$ sufficiently small. By construction $\tau(\Delta_t) = D$. In particular $\tau(0, 0, \zeta)$ is realized as the modulus of some origami torus. This completes the proof of the Universality Theorem.

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